

Demographic Forecasting

Gary King
Harvard University

Joint work with Federico Girosi (RAND)
with contributions from Kevin Quinn and Gregory Wawro

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- Approach: Formalizing **qualitative** insights in **quantitative** models

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- Better ways of creating Bayesian priors
- Produces forecasts and farcasts using considerably more information

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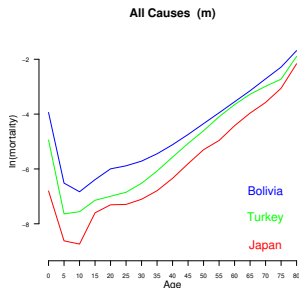
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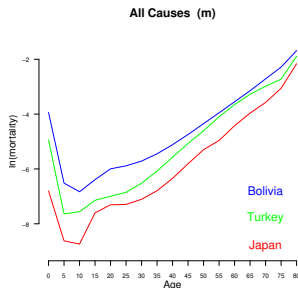
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Existing Method 1: Parameterize the Age Profile

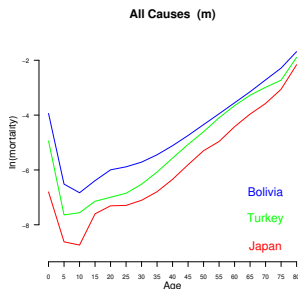


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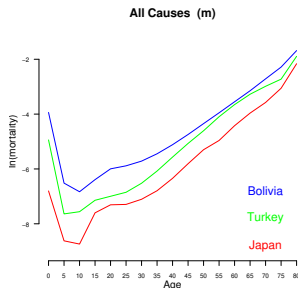
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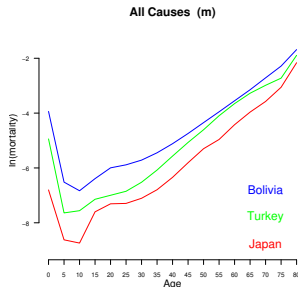
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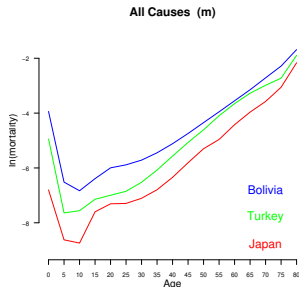
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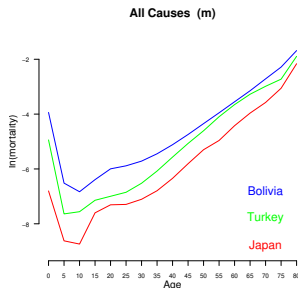
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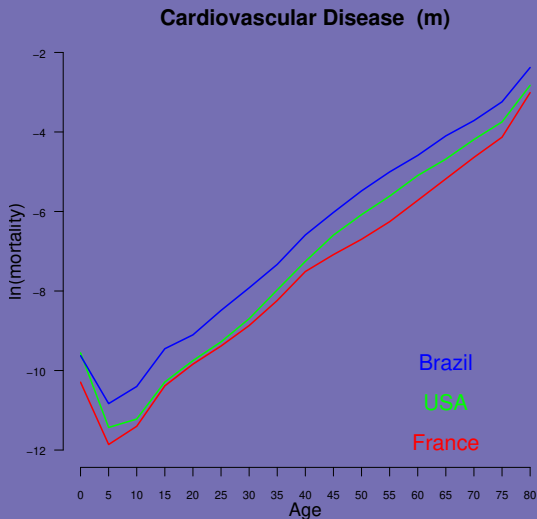
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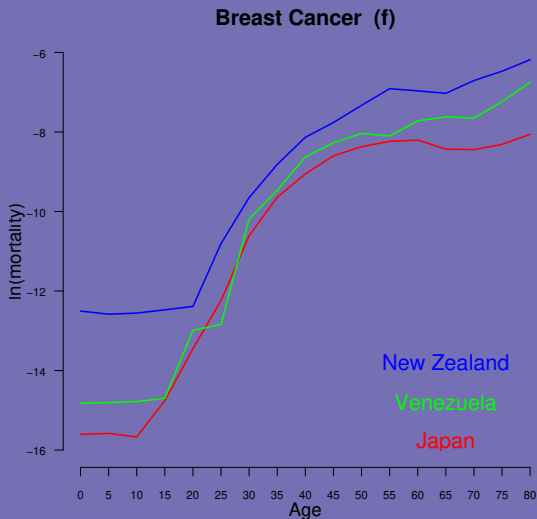


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 - reduces 17 age-specific mortality rates to 2 parameters (intercept and slope)
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- Dozens of more general functional forms proposed
- **But does it fit anything else?**

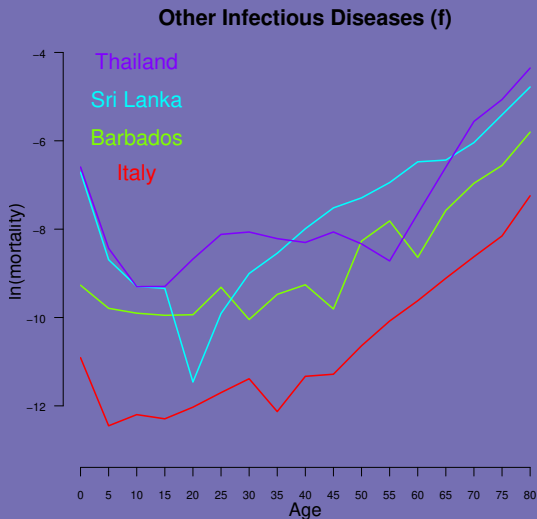
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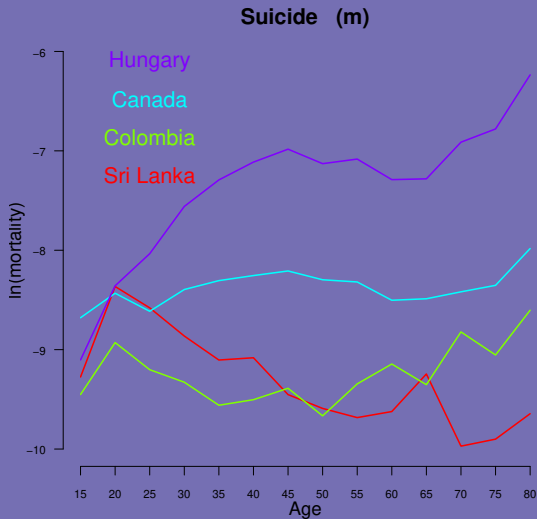
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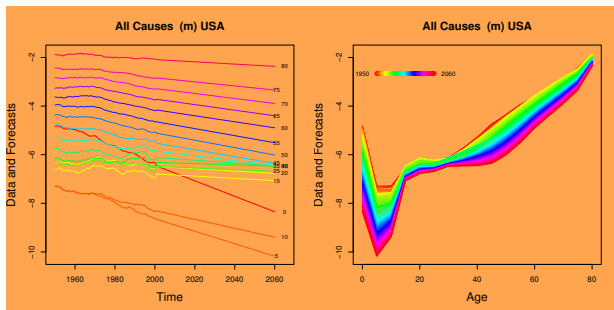
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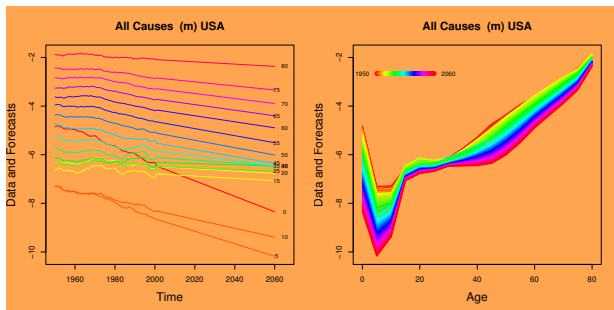
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- Also: Method ignores covariate information; the leading current method (McNown-Rogers) not replicable

Existing Method 2: Deterministic Projections

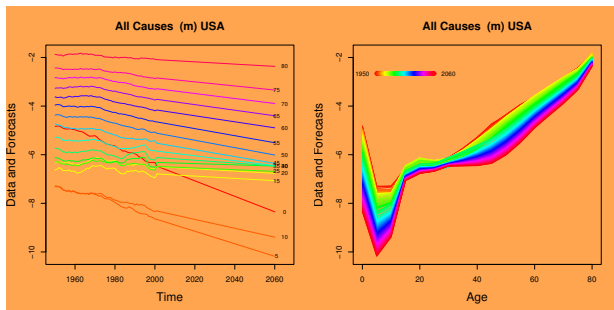


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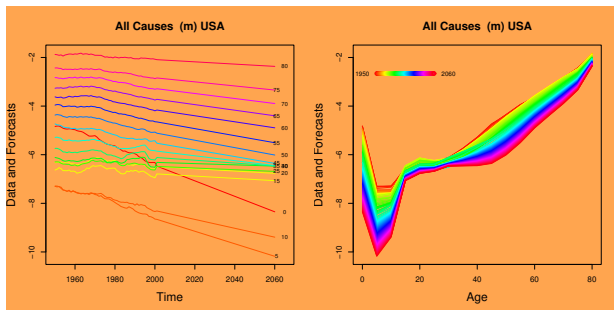
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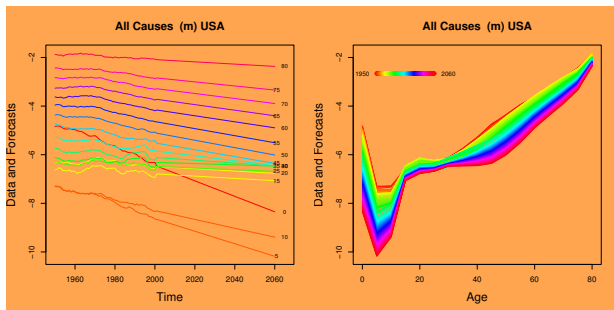
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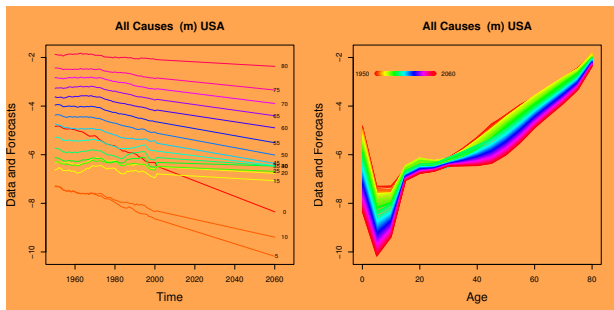
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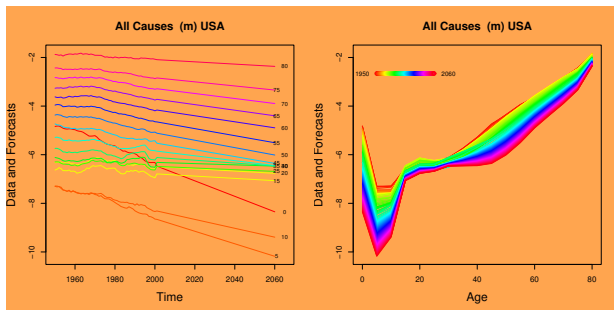
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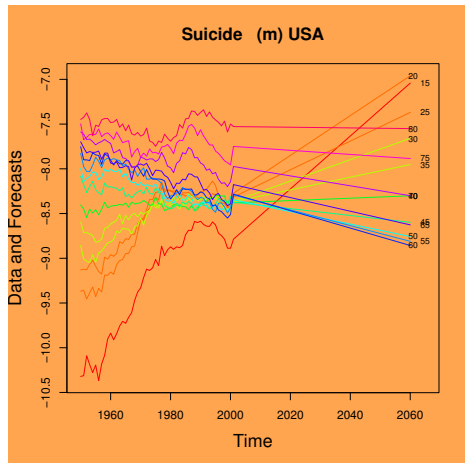
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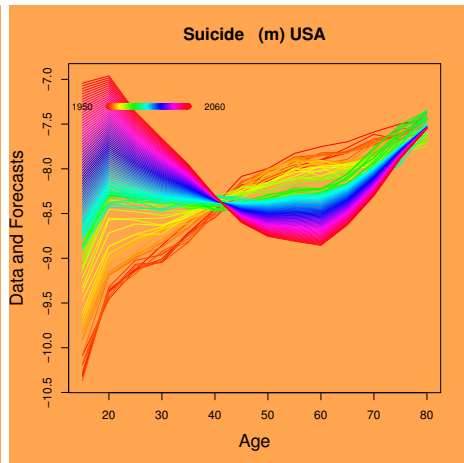
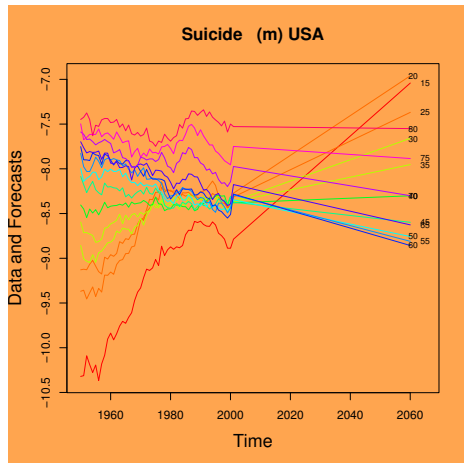
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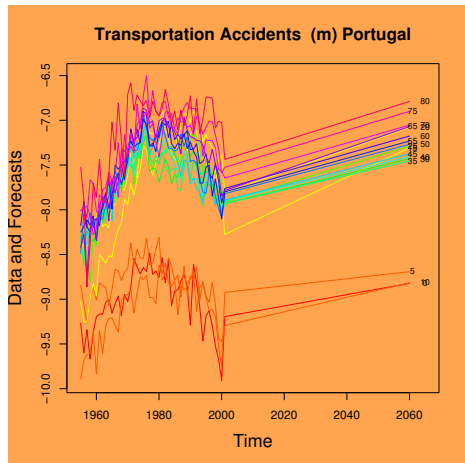
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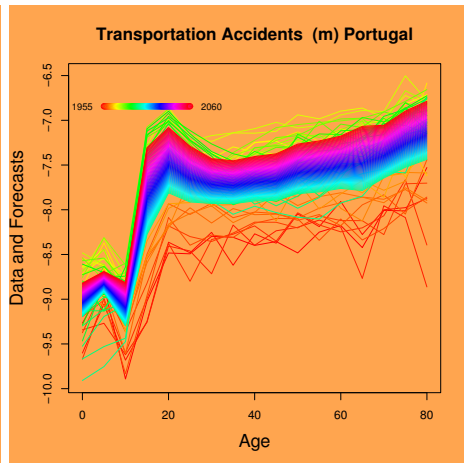
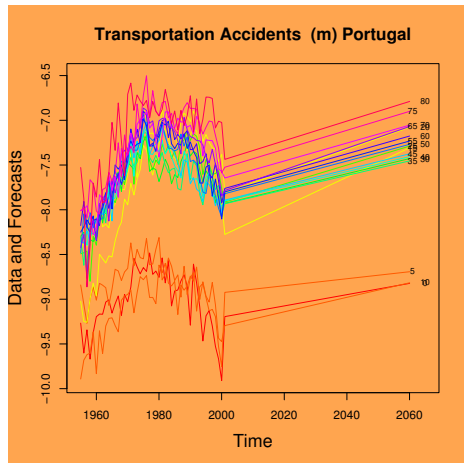
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 - considerable information lost (due to no pooling over ages)

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- Cannot estimate equation by equation (variance is too large);
- Pool over countries: $\boldsymbol{\beta}_{ca} \Rightarrow \boldsymbol{\beta}_a$
- Properties:
 - Small variance (due to large n)
 - large biases (due to restrictive pooling over countries),
 - considerable information lost (due to no pooling over ages)
 - same covariates required in all cross-sections

Partial Pooling via a Bayesian Hierarchical Approach

- Likelihood for equation-by-equation least squares:

$$\mathcal{P}(m \mid \beta_i, \sigma_i) = \prod_t \mathcal{N}(m_{it} \mid \mathbf{z}_{it}\beta_i, \sigma_i^2)$$

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- The easy part: *easy-to-use software* to implement everything we discuss today.

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Natural choice for the prior:

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- Extensive trial-and-error runs, yielded no useful parameter values.

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 - 3 In the subspace, we can invert $\mu = \mathbf{Z}\beta$ as $\beta = (\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mu$, giving:

$$\mathcal{P}(\beta \mid \theta) \propto \exp\left(-\frac{1}{2}H[\mu, \theta]\right) = \exp\left(-\frac{1}{2}H[\mathbf{Z}\beta, \theta]\right)$$

the same prior on μ , expressed as a function of β (with constant Jacobian).

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- We start with one-dimensional $P(\mu_{cat})$, and treat it as the multidimensional $P(\beta_{ca})$, constant in all directions but $Z_{cat}\beta_{ca}$.

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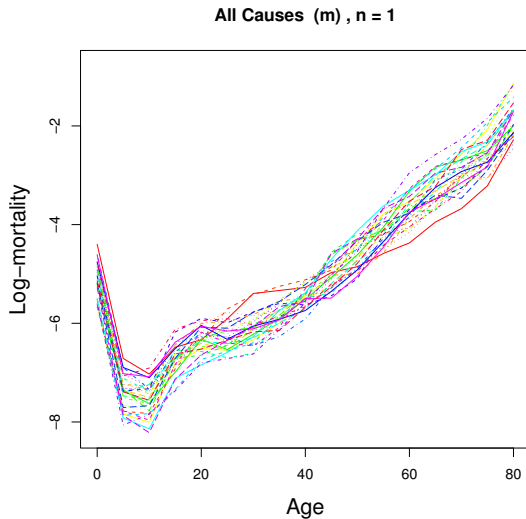
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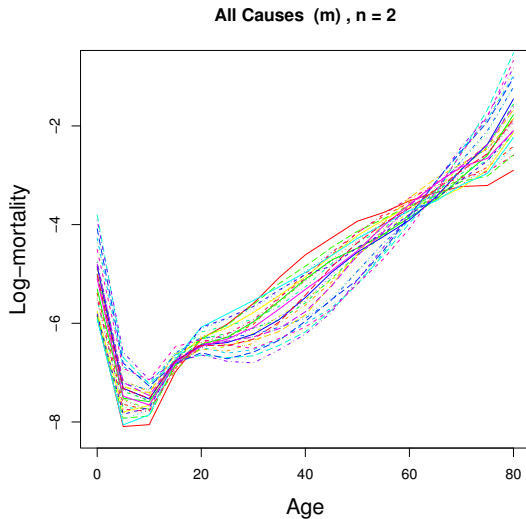
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- Different age groups can have different covariates: the matrices $\mathbf{C}_{aa'} \equiv \frac{1}{T} \mathbf{Z}'_a \mathbf{Z}_{a'}$ are rectangular ($d_a \times d_{a'}$).

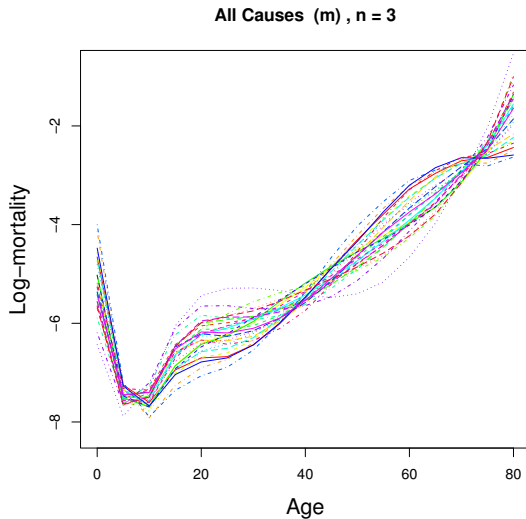
Samples From Age Prior



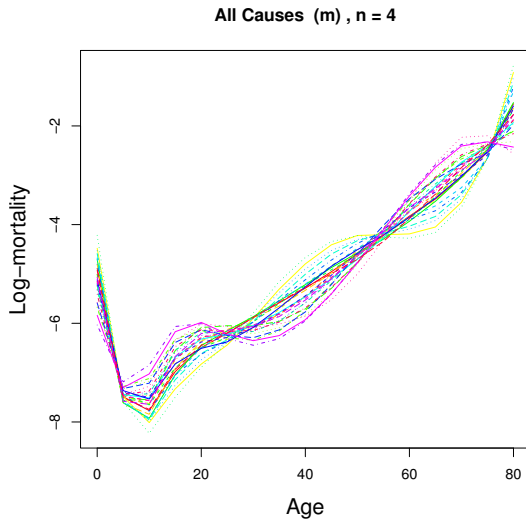
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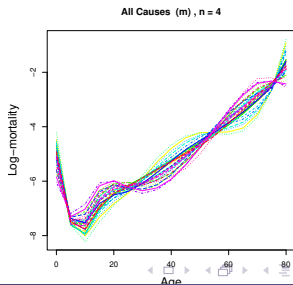
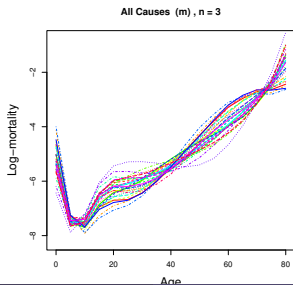
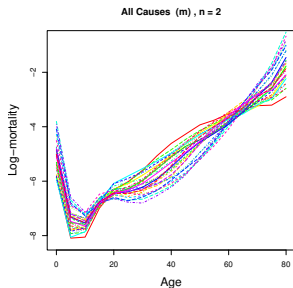
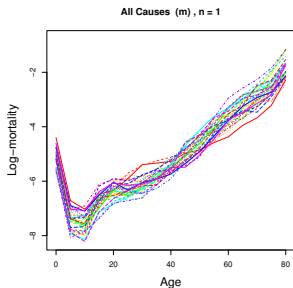
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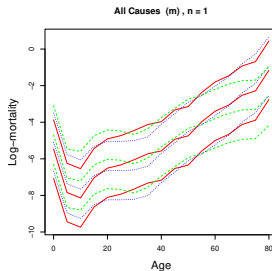
Formalizing (Prior) Indifference

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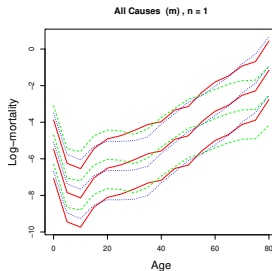
Level indifference



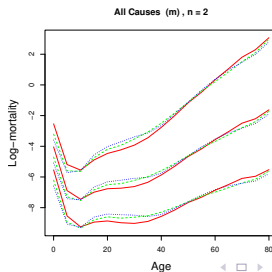
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Level and slope indifference



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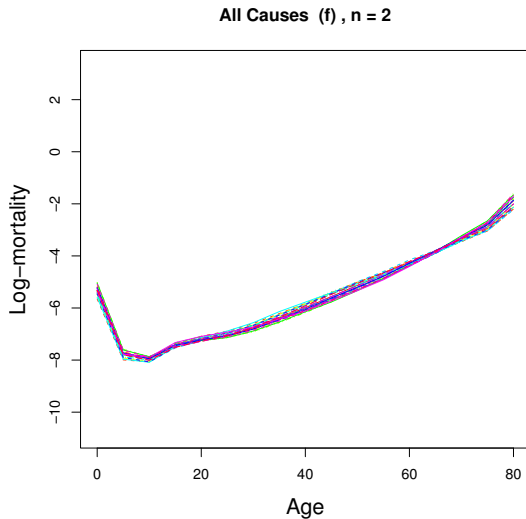
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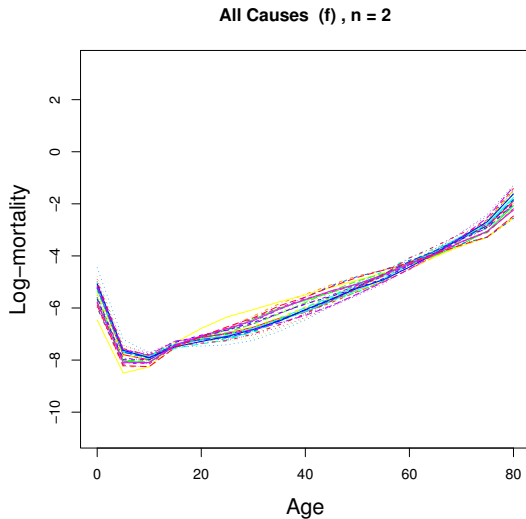
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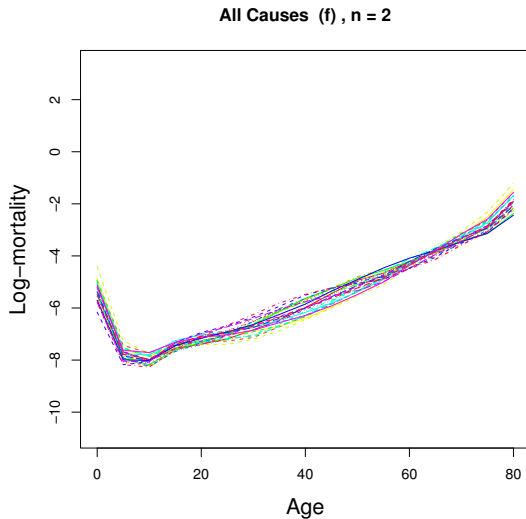
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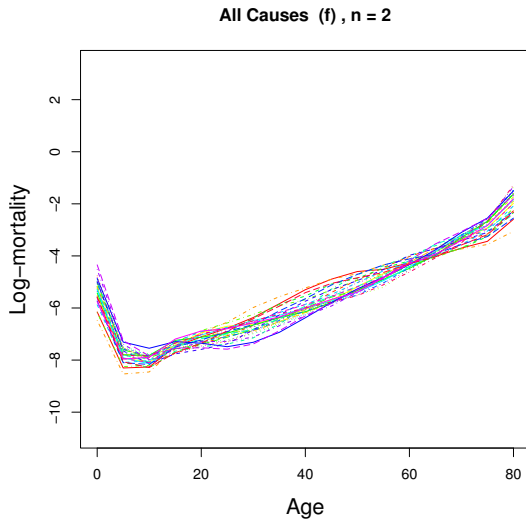
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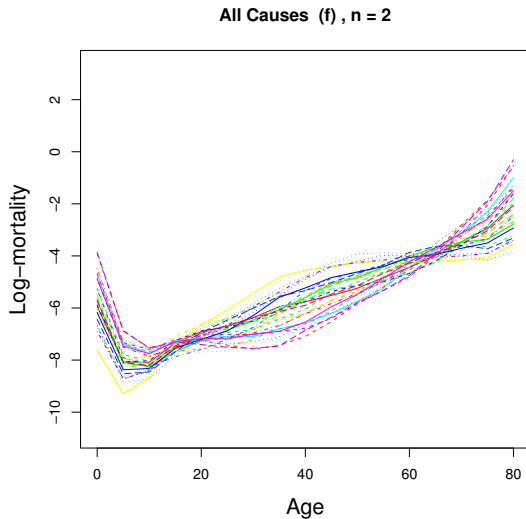
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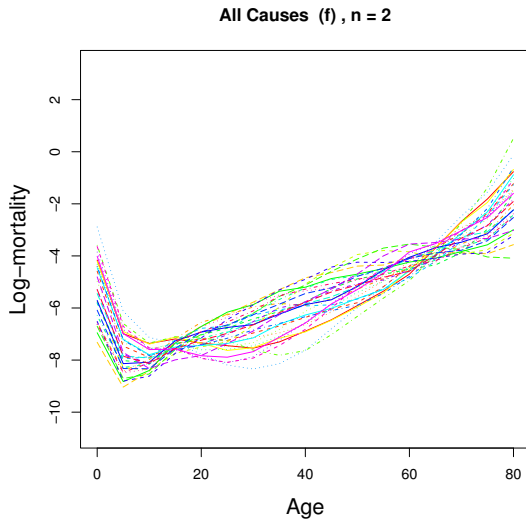
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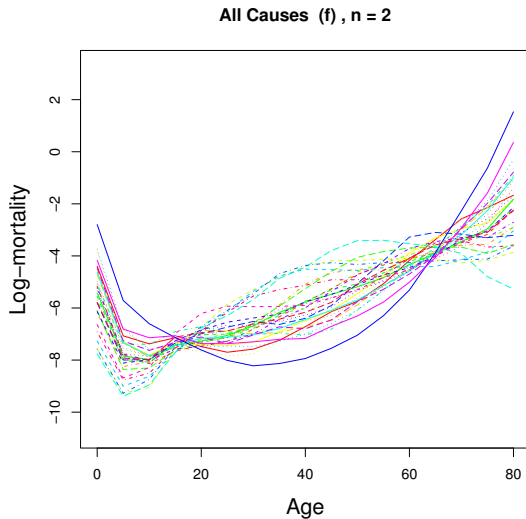
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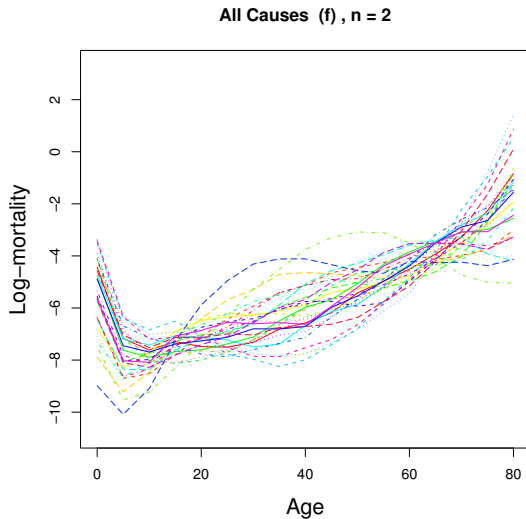
Samples from Age Prior



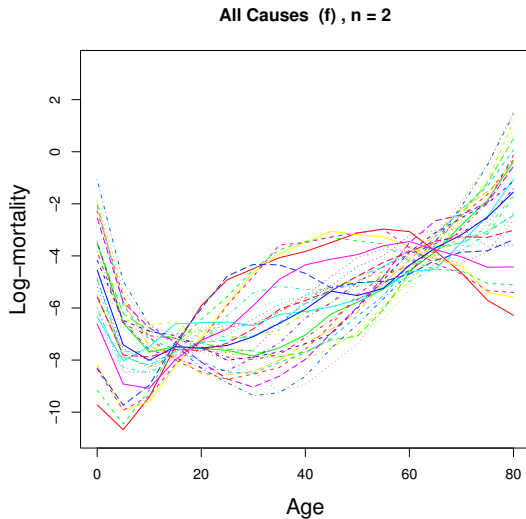
Samples from Age Prior



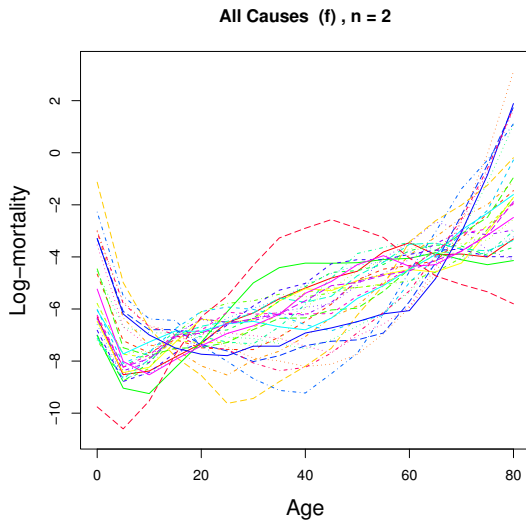
Samples from Age Prior



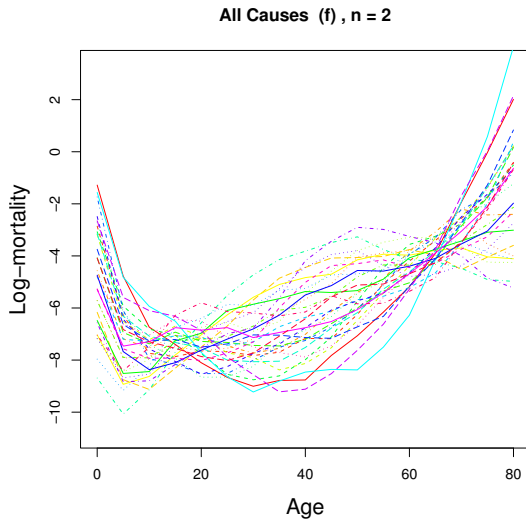
Samples from Age Prior



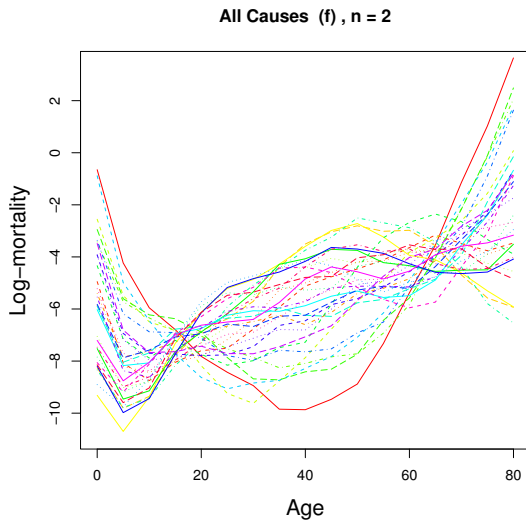
Samples from Age Prior



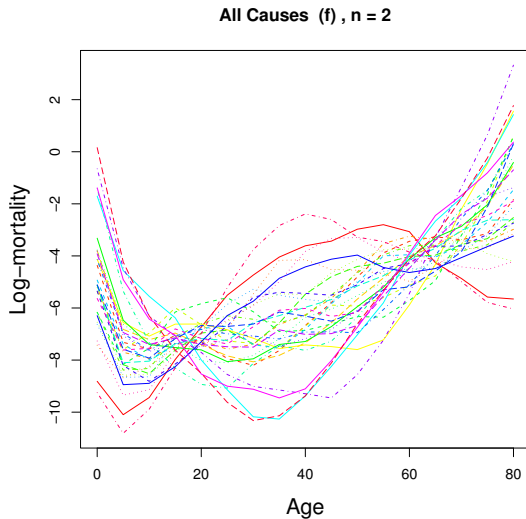
Samples from Age Prior



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Generalizations

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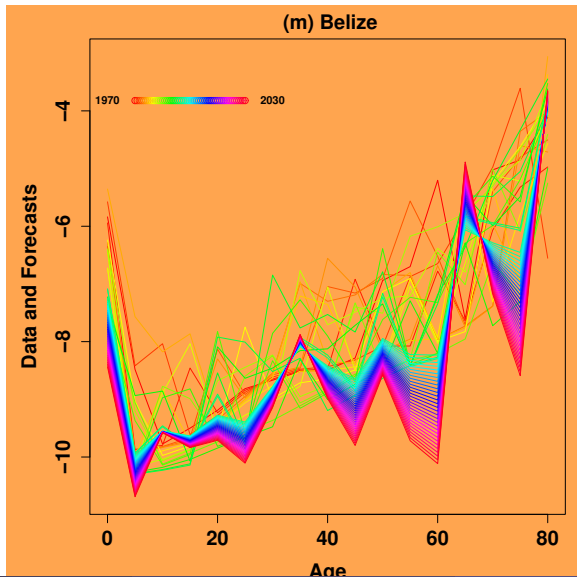
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- The mathematical form for *all* these (separately or together) turns out to be the same:

$$\mathcal{P}(\boldsymbol{\beta} \mid \theta) \propto \exp \left(-\frac{\theta}{2} \sum_{ij} W_{ij} \boldsymbol{\beta}'_i \mathbf{C}_{ij} \boldsymbol{\beta}_j \right), \quad \mathbf{C}_{aa'} \equiv \frac{1}{T} \mathbf{Z}_a \mathbf{Z}_{a'}$$

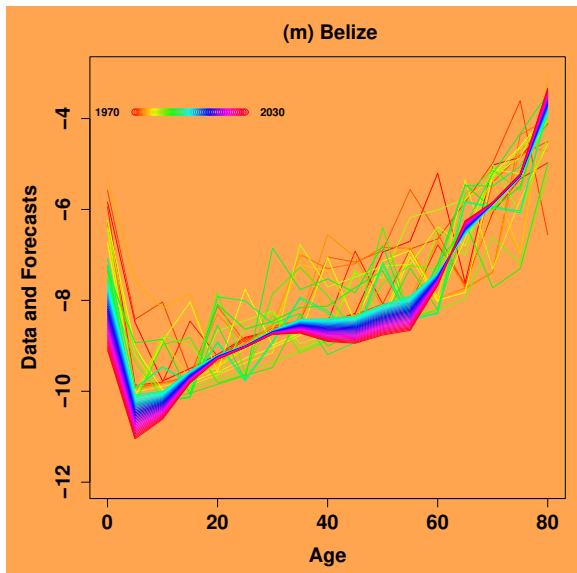
Mortality from Respiratory Infections, Males

Least Squares



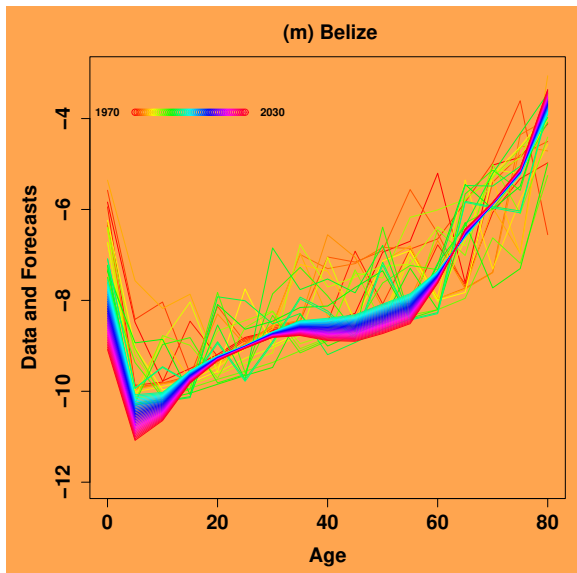
Mortality from Respiratory Infections, males, $\sigma = 2.00$

Smoothing over Age Groups



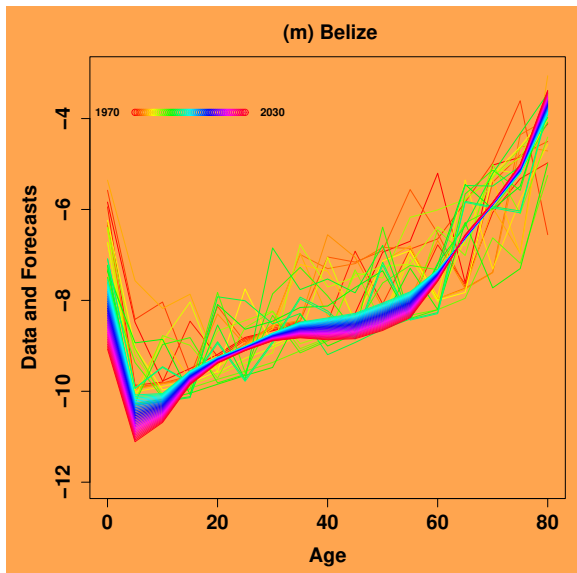
Mortality from Respiratory Infections, males, $\sigma = 1.51$

Smoothing over Age Groups



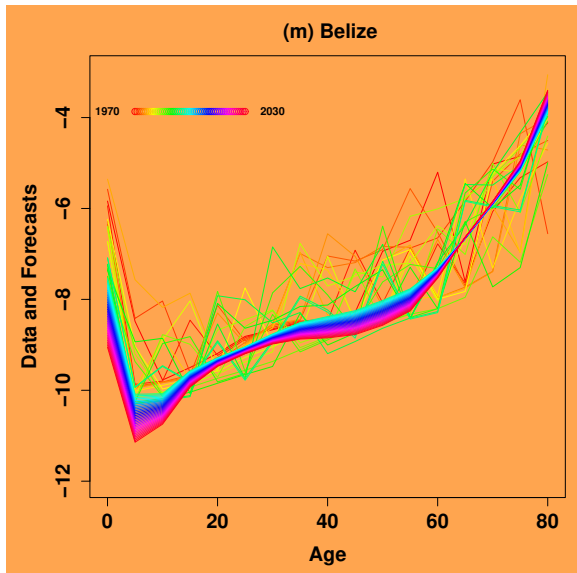
Mortality from Respiratory Infections, males, $\sigma = 1.15$

Smoothing over Age Groups



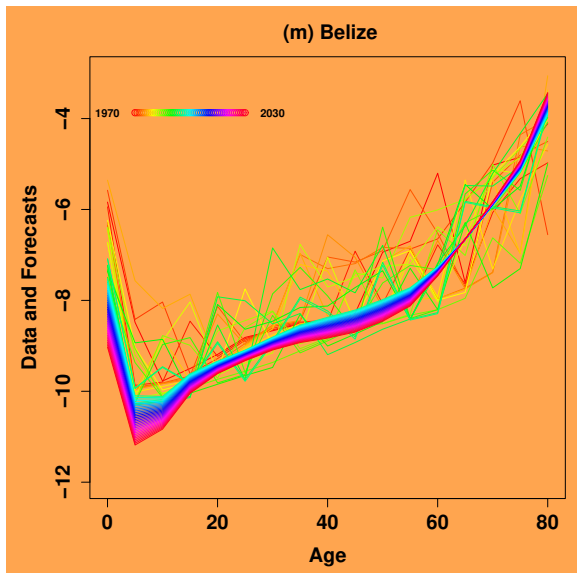
Mortality from Respiratory Infections, males, $\sigma = 0.87$

Smoothing over Age Groups



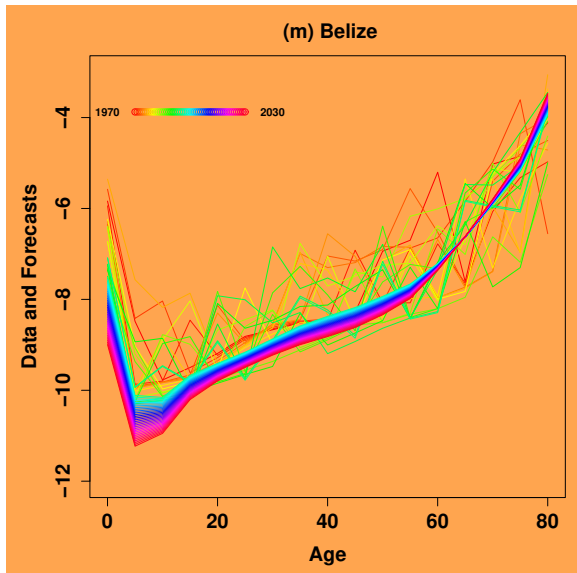
Mortality from Respiratory Infections, males, $\sigma = 0.66$

Smoothing over Age Groups



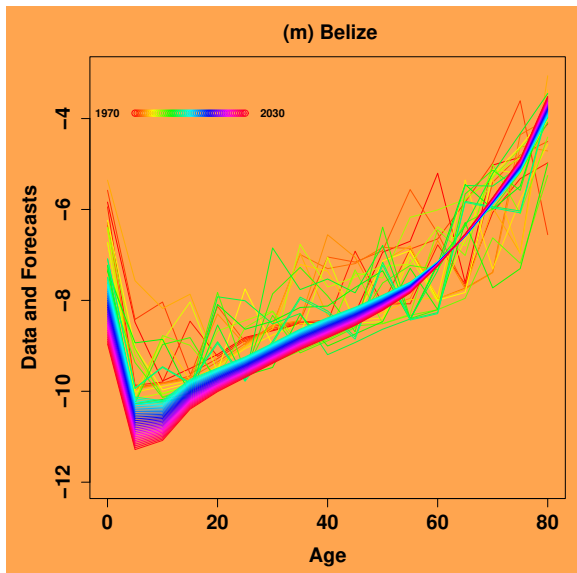
Mortality from Respiratory Infections, males, $\sigma = 0.50$

Smoothing over Age Groups



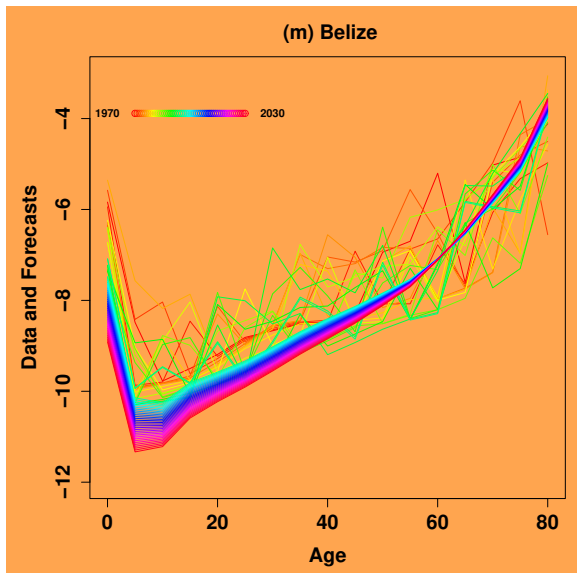
Mortality from Respiratory Infections, males, $\sigma = 0.38$

Smoothing over Age Groups



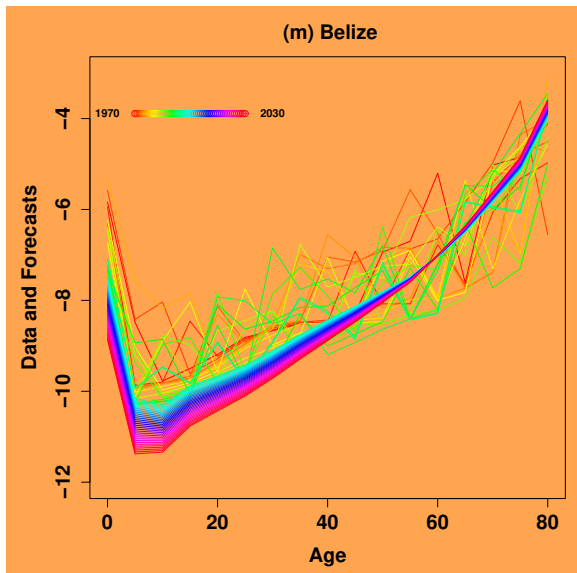
Mortality from Respiratory Infections, males, $\sigma = 0.28$

Smoothing over Age Groups



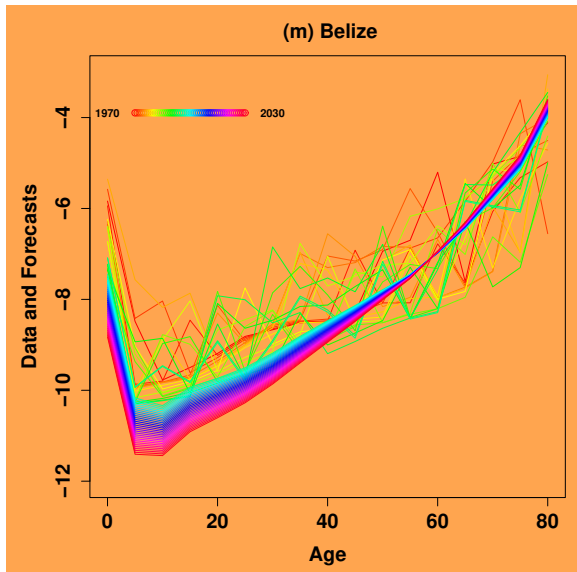
Mortality from Respiratory Infections, males, $\sigma = 0.21$

Smoothing over Age Groups



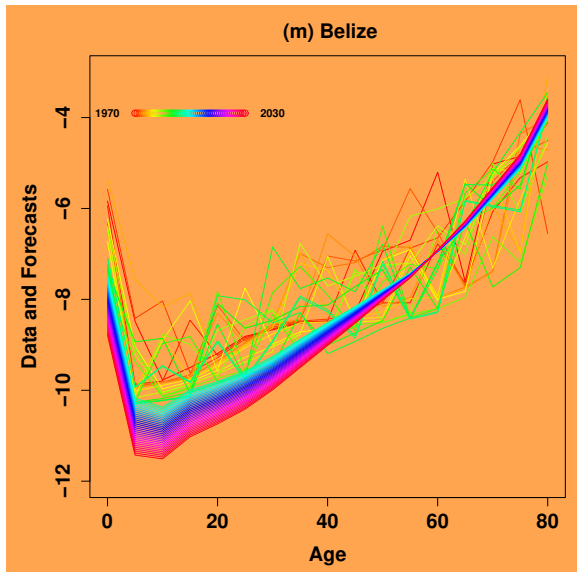
Mortality from Respiratory Infections, males, $\sigma = 0.16$

Smoothing over Age Groups



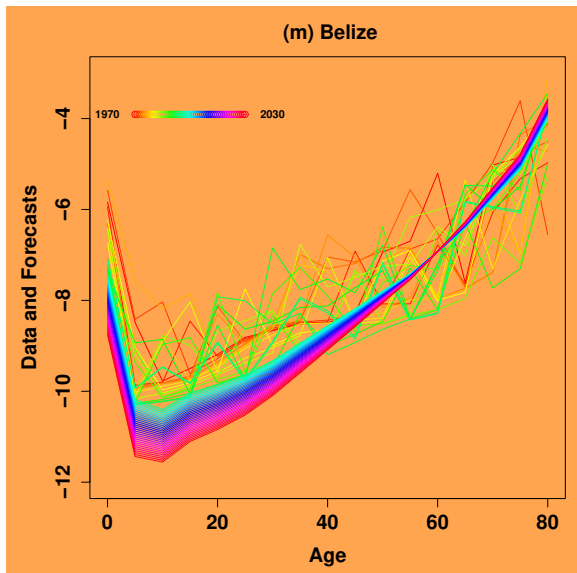
Mortality from Respiratory Infections, males, $\sigma = 0.12$

Smoothing over Age Groups



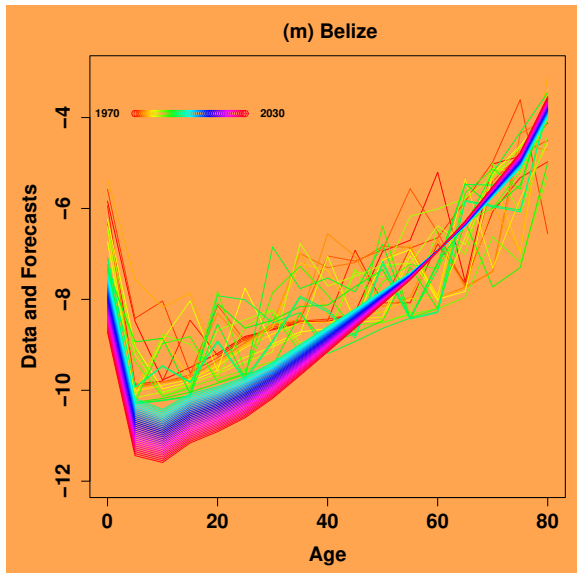
Mortality from Respiratory Infections, males, $\sigma = 0.09$

Smoothing over Age Groups



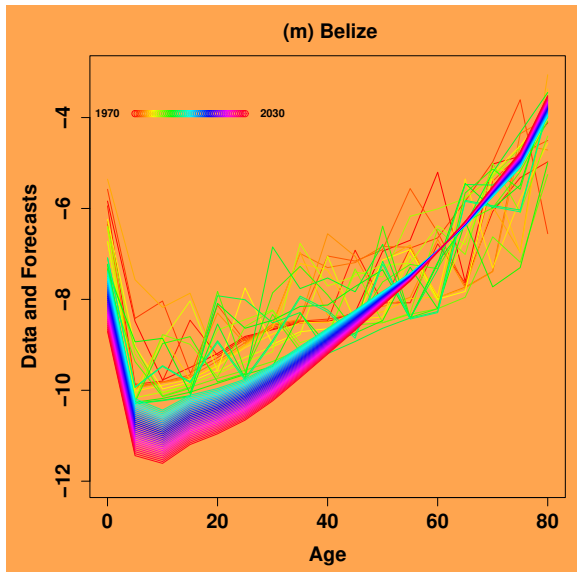
Mortality from Respiratory Infections, males, $\sigma = 0.07$

Smoothing over Age Groups



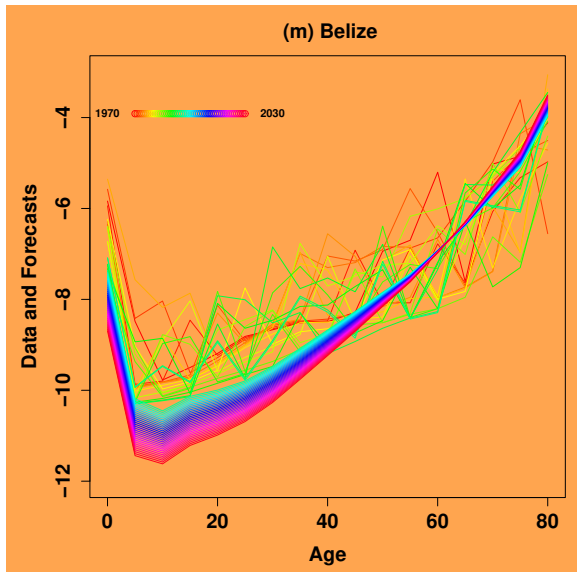
Mortality from Respiratory Infections, males, $\sigma = 0.05$

Smoothing over Age Groups



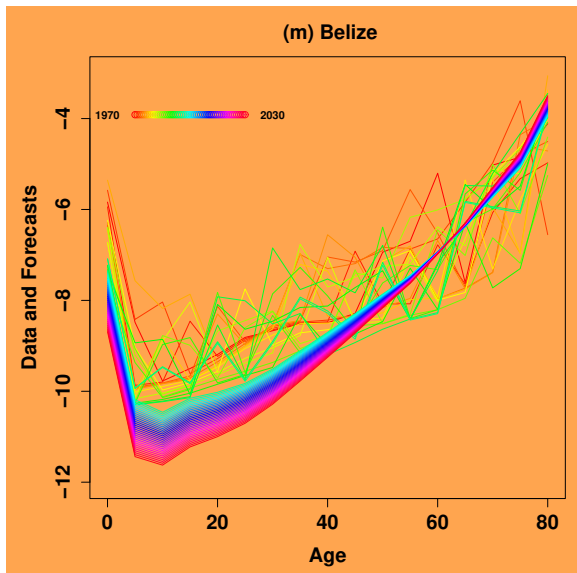
Mortality from Respiratory Infections, males, $\sigma = 0.04$

Smoothing over Age Groups



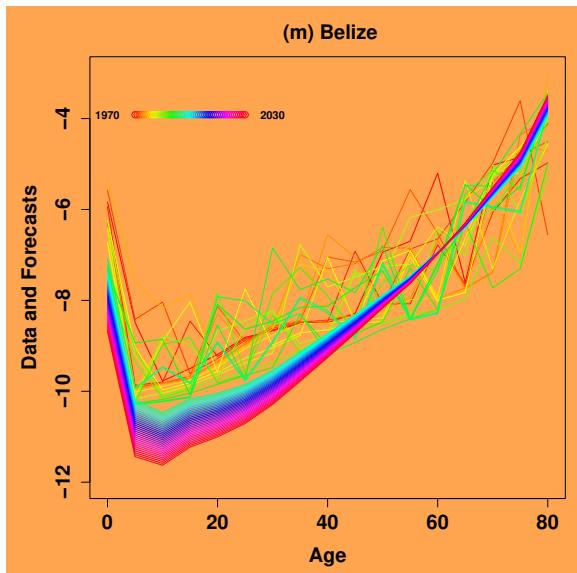
Mortality from Respiratory Infections, males, $\sigma = 0.03$

Smoothing over Age Groups



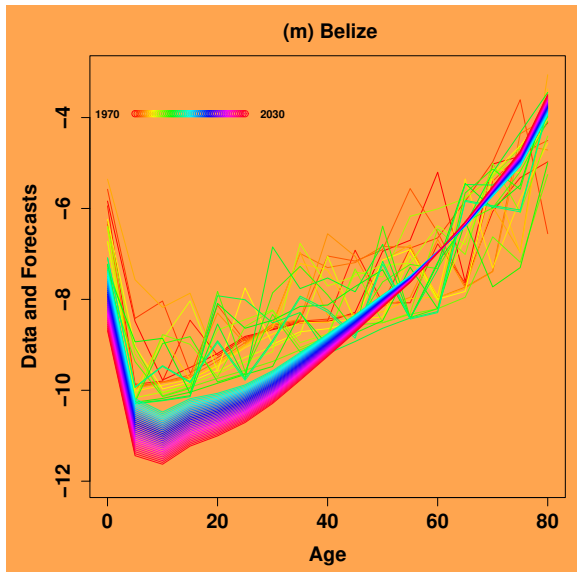
Mortality from Respiratory Infections, males, $\sigma = 0.02$

Smoothing over Age Groups



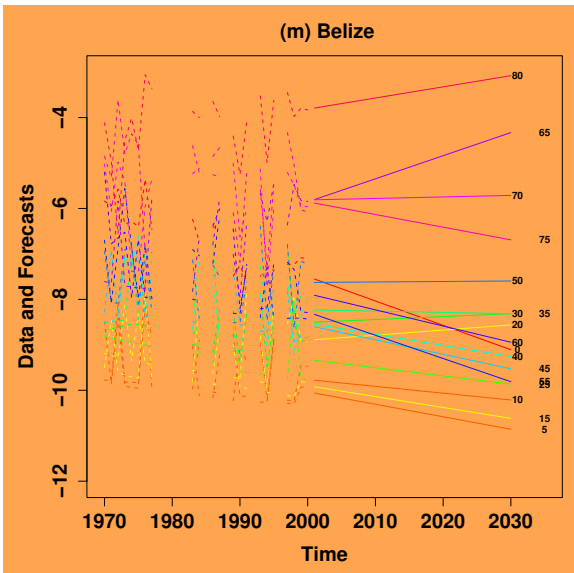
Mortality from Respiratory Infections, males, $\sigma = 0.01$

Smoothing over Age Groups



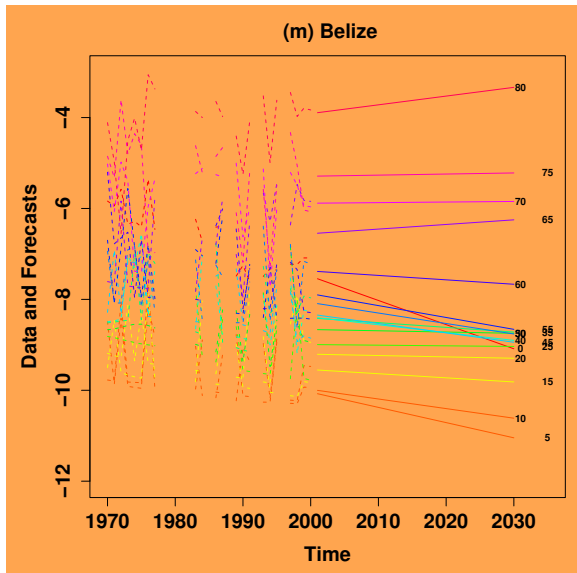
Mortality from Respiratory Infections, males

Least Squares



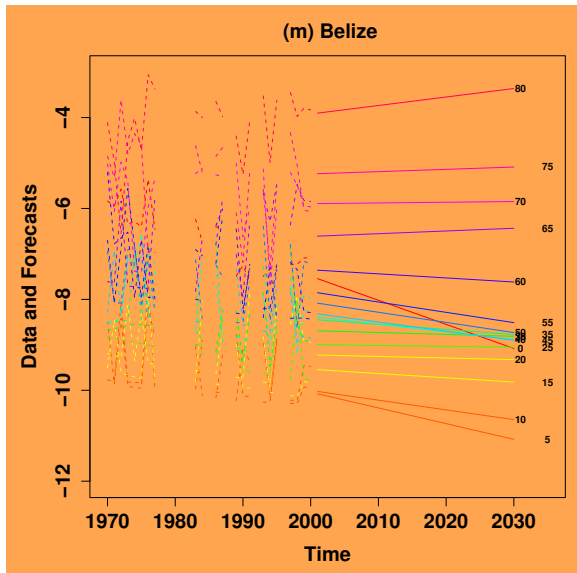
Mortality from Respiratory Infections, males, $\sigma = 2.00$

Smoothing over Age Groups



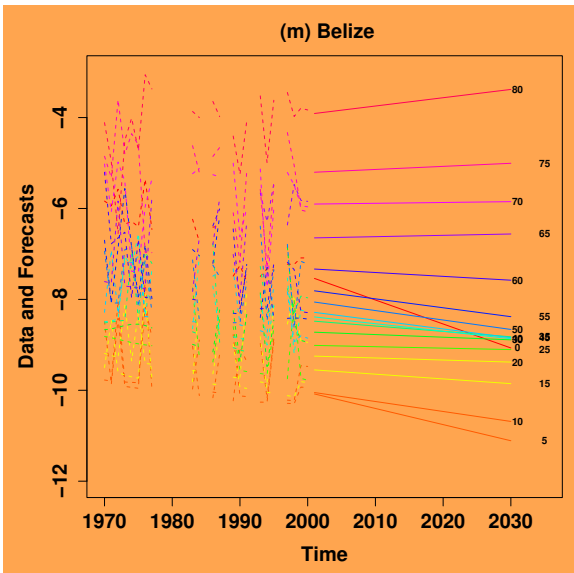
Mortality from Respiratory Infections, males, $\sigma = 1.51$

Smoothing over Age Groups



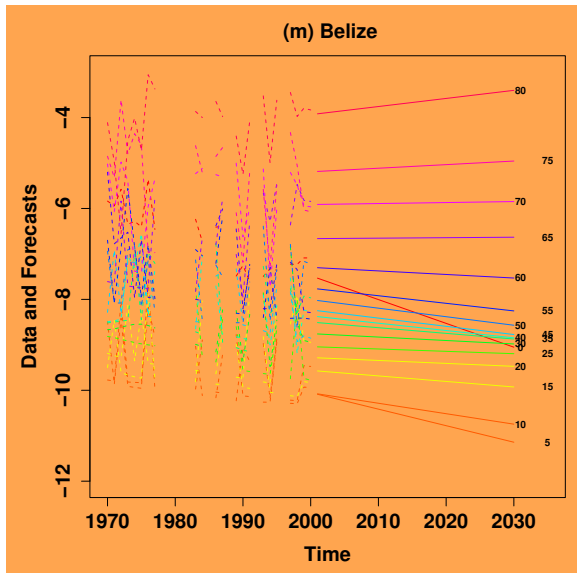
Mortality from Respiratory Infections, males, $\sigma = 1.15$

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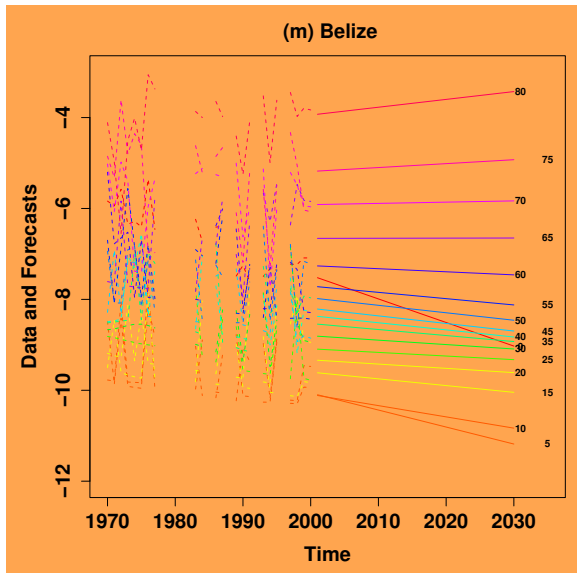
Mortality from Respiratory Infections, males, $\sigma = 0.87$

Smoothing over Age Groups



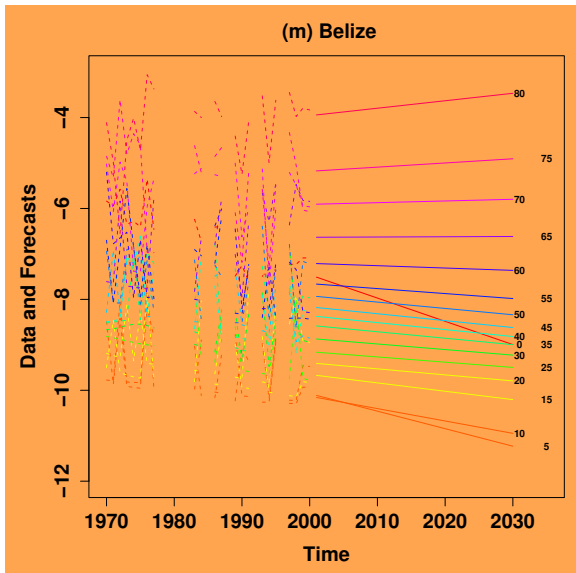
Mortality from Respiratory Infections, males, $\sigma = 0.66$

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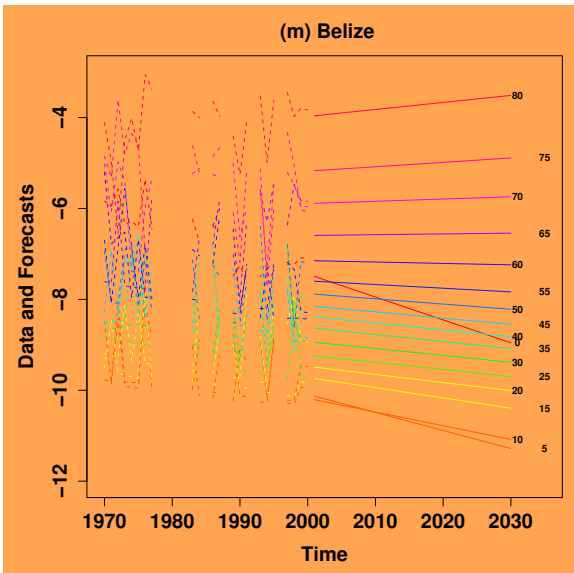
Mortality from Respiratory Infections, males, $\sigma = 0.50$

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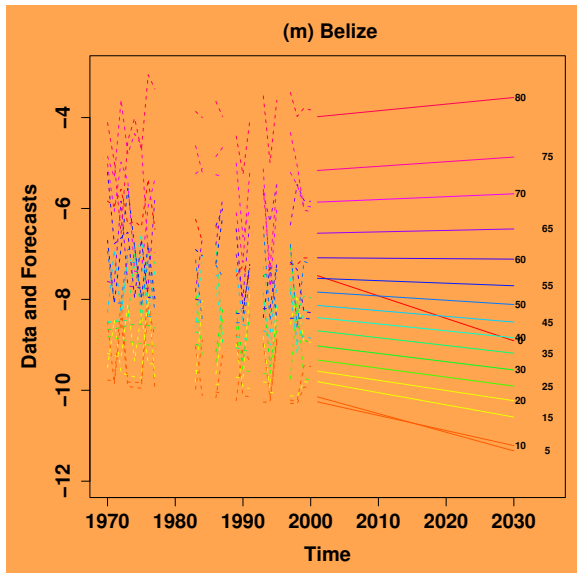
Mortality from Respiratory Infections, males, $\sigma = 0.38$

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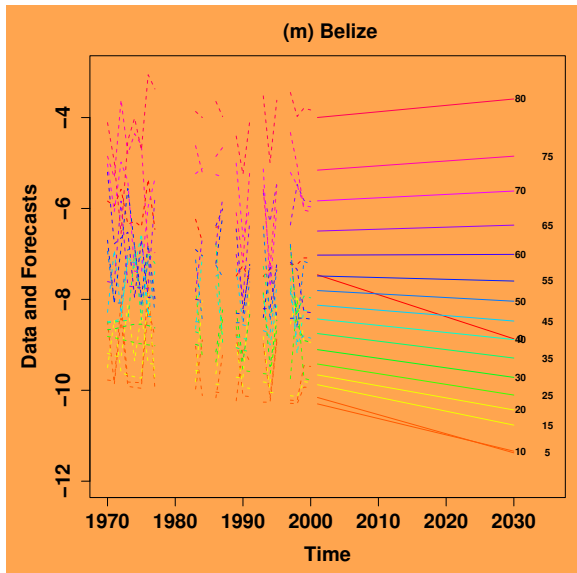
Mortality from Respiratory Infections, males, $\sigma = 0.28$

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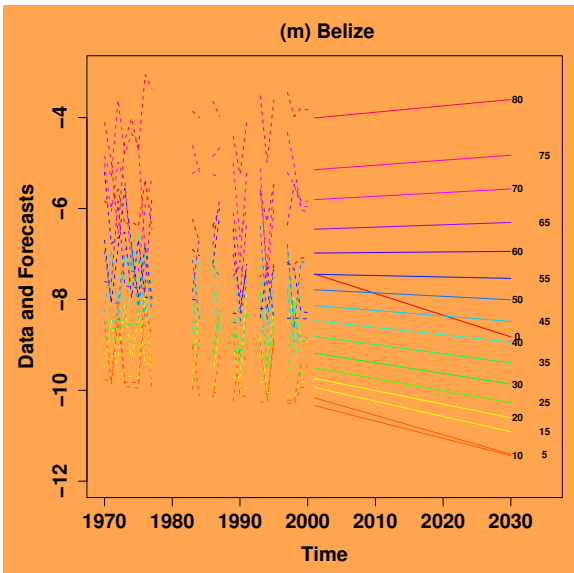
Mortality from Respiratory Infections, males, $\sigma = 0.21$

Smoothing over Age Groups



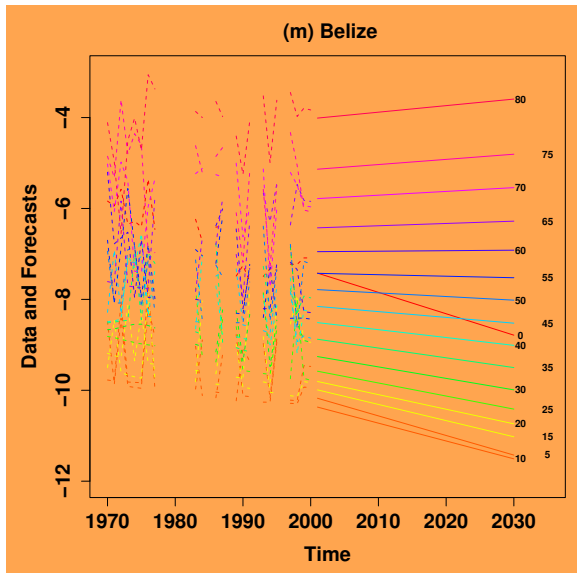
Mortality from Respiratory Infections, males, $\sigma = 0.16$

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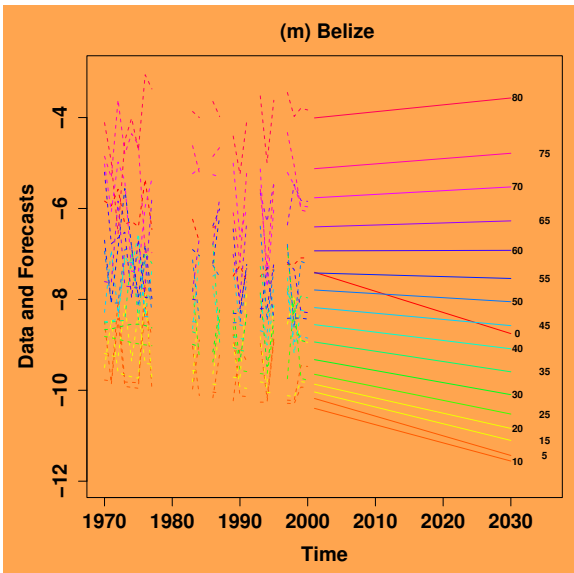
Mortality from Respiratory Infections, males, $\sigma = 0.12$

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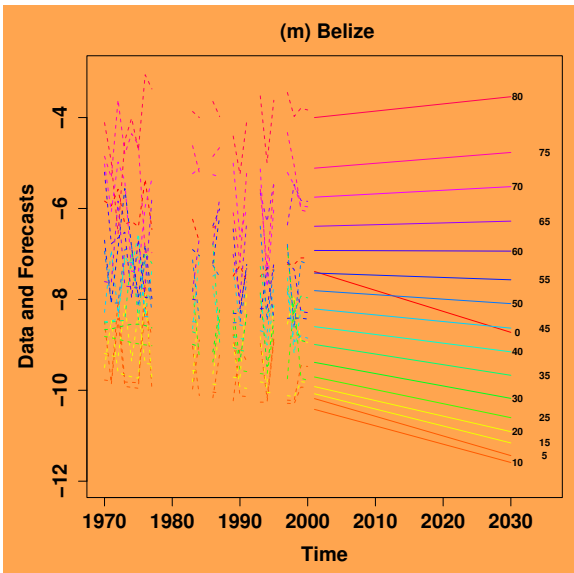
Mortality from Respiratory Infections, males, $\sigma = 0.09$

Smoothing over Age Groups



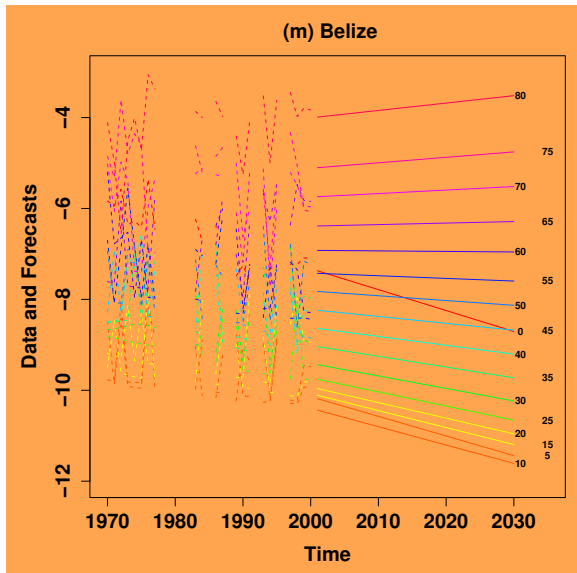
Mortality from Respiratory Infections, males, $\sigma = 0.07$

Smoothing over Age Groups



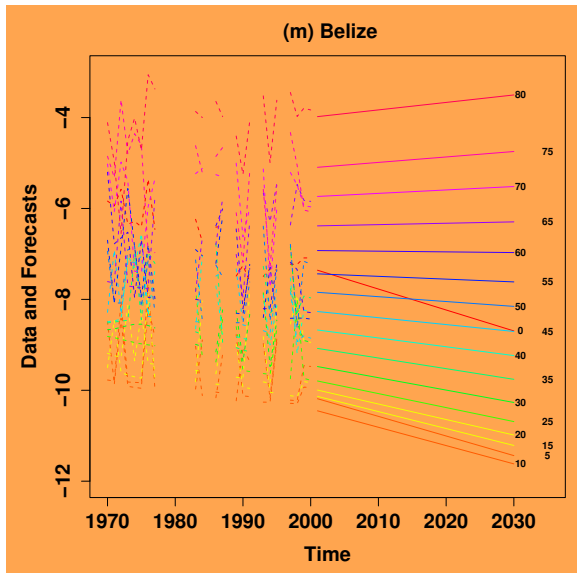
Mortality from Respiratory Infections, males, $\sigma = 0.05$

Smoothing over Age Groups



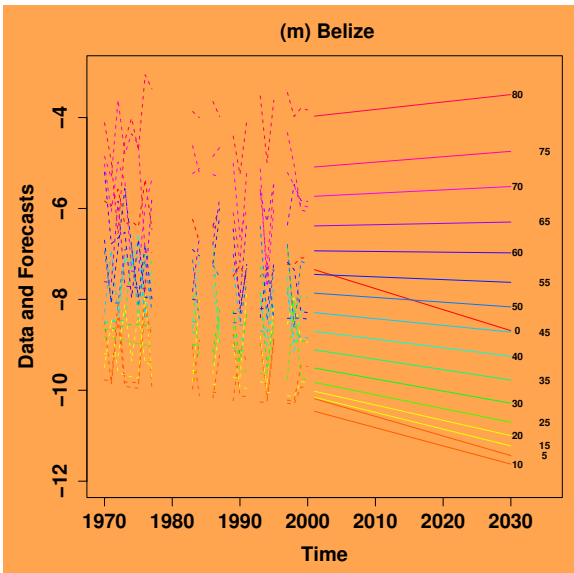
Mortality from Respiratory Infections, males, $\sigma = 0.04$

Smoothing over Age Groups



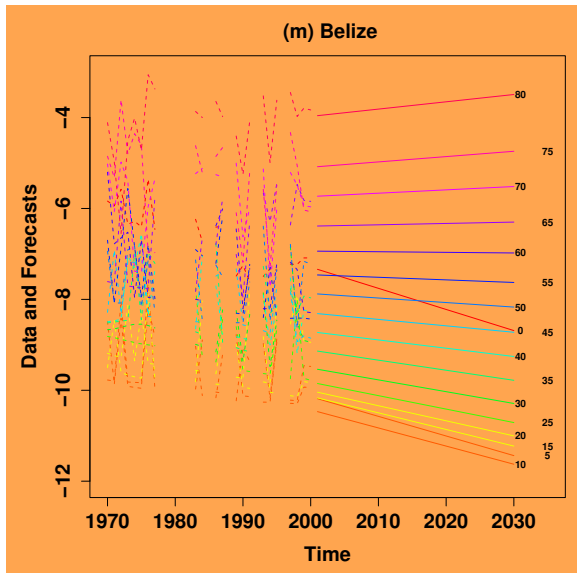
Mortality from Respiratory Infections, males, $\sigma = 0.03$

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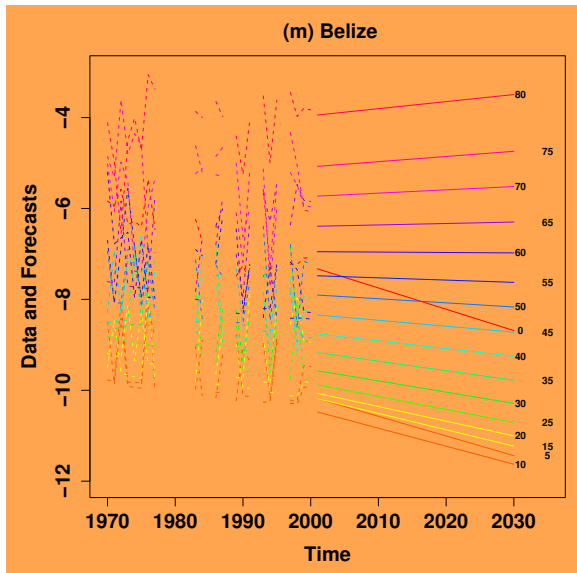
Mortality from Respiratory Infections, males, $\sigma = 0.02$

Smoothing over Age Groups



Mortality from Respiratory Infections, males, $\sigma = 0.01$

Smoothing over Age Groups



Smoothing Trends over Age Groups

Smoothing Trends over Age Groups

Log-mortality in Belize males from respiratory infections

Smoothing Trends over Age Groups

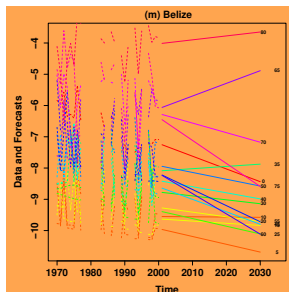
Log-mortality in Belize males from respiratory infections

Least Squares

Smoothing Trends over Age Groups

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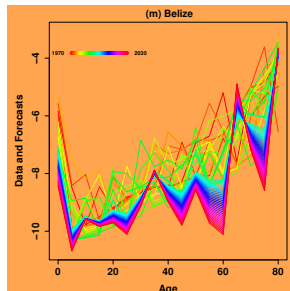
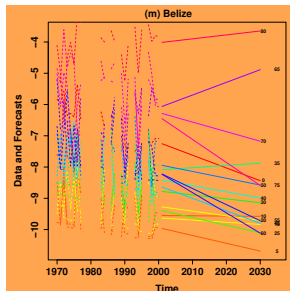
Least Squares



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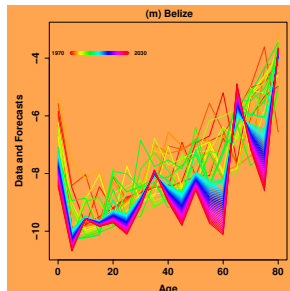
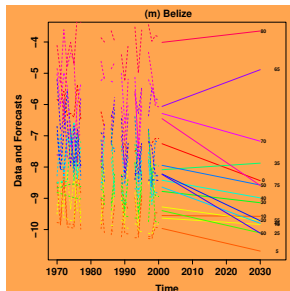
Least Squares



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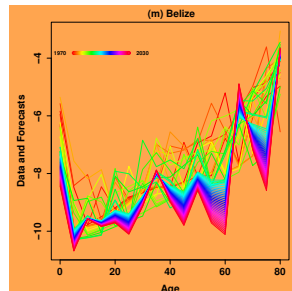
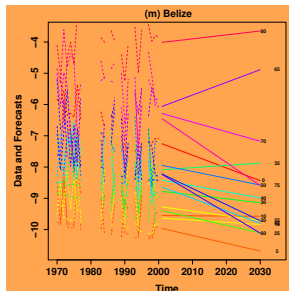


Smoothing
Age Groups

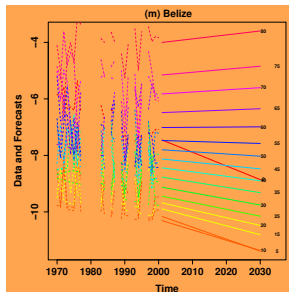
Smoothing Trends over Age Groups

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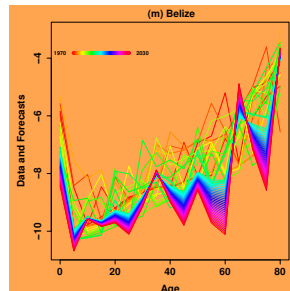
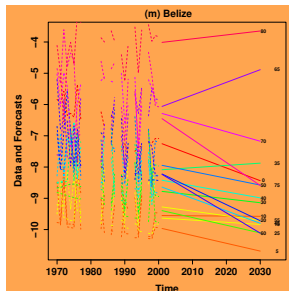
Smoothing
Age Groups



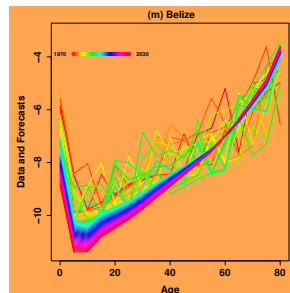
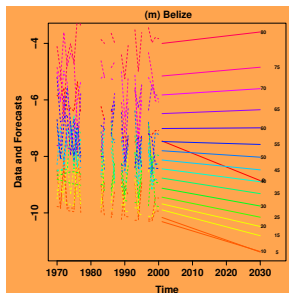
Smoothing Trends over Age Groups

Log-mortality in Belize males from respiratory infections

Least Squares



Smoothing
Age Groups



Smoothing Trends over Age Groups and Time

Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

Smoothing Trends over Age Groups and Time

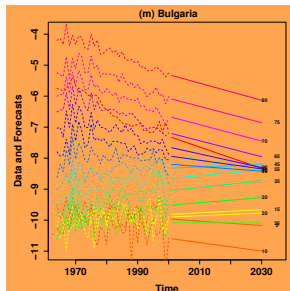
Log-Mortality in Bulgarian males from respiratory infections

Least Squares

Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

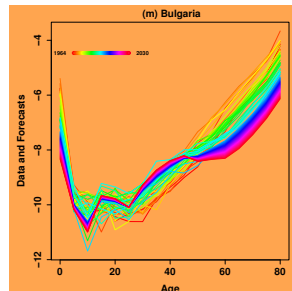
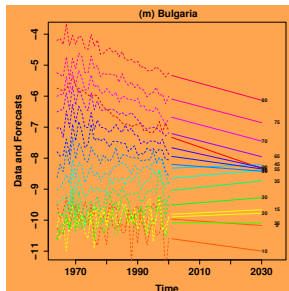
Least Squares



Smoothing Trends over Age Groups and Time

Log-Mortality in Bulgarian males from respiratory infections

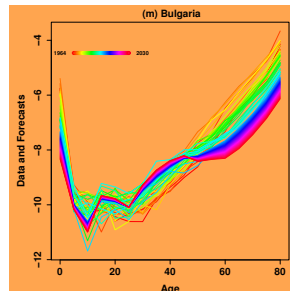
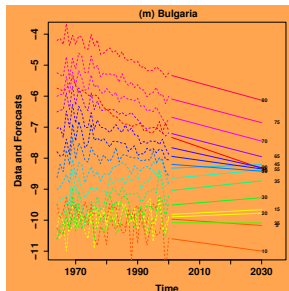
Least Squares



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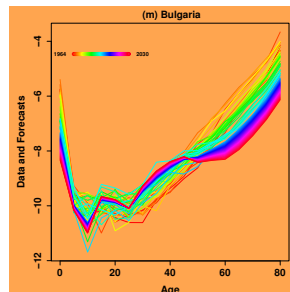
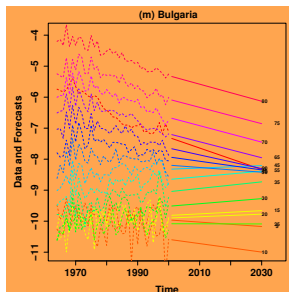


Smoothing
Age and Time

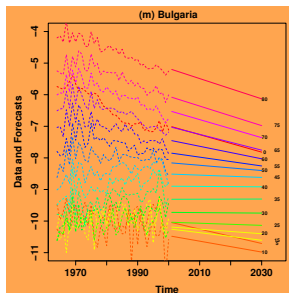
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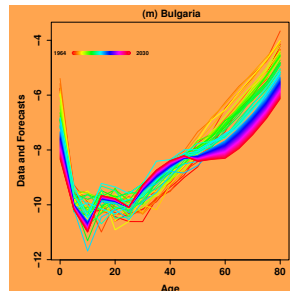
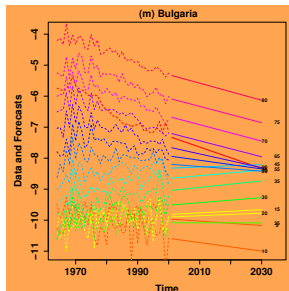
Smoothing
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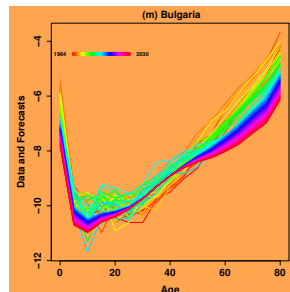
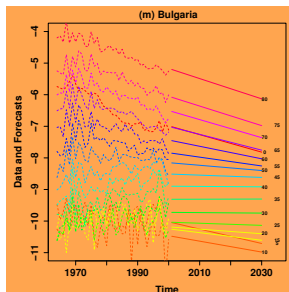
Smoothing Trends over Age Groups and Time

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Smoothing
Age and Time



Using Covariates (GDP, tobacco, trend, log trend)

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Lung cancer in Korean Males

Using Covariates (GDP, tobacco, trend, log trend)

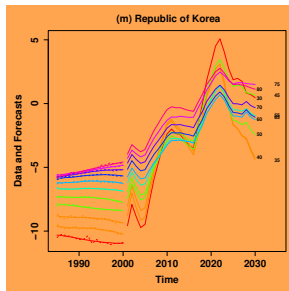
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Least Squares

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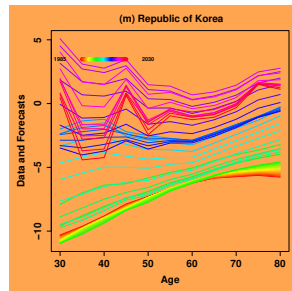
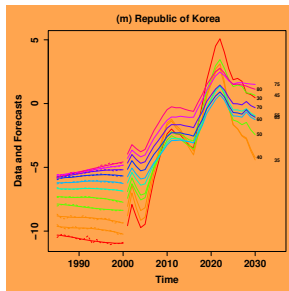
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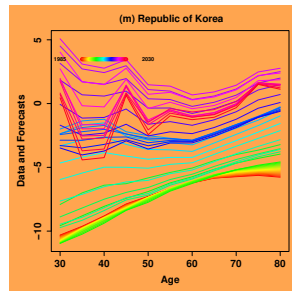
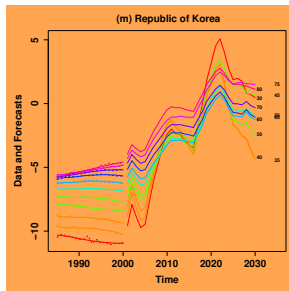
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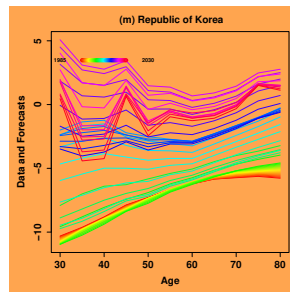
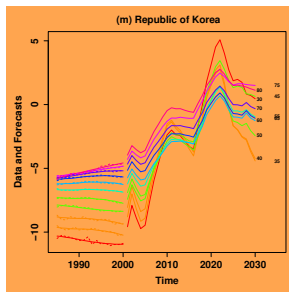


Smooth over age,
time, age/time

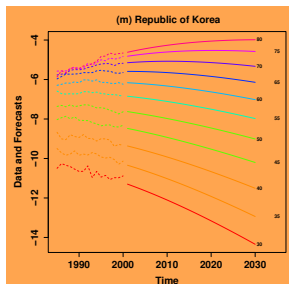
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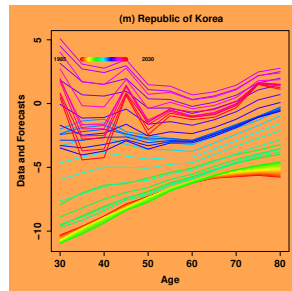
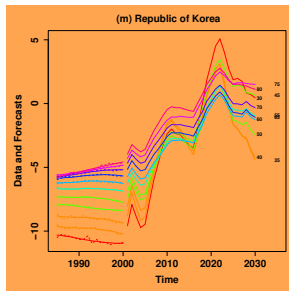
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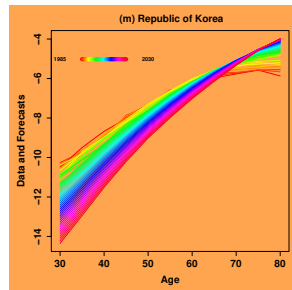
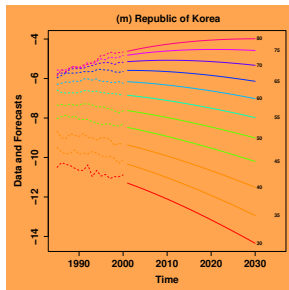
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Using Covariates (GDP, tobacco, trend, log trend)

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Lung cancer in Males, Singapore

Using Covariates (GDP, tobacco, trend, log trend)

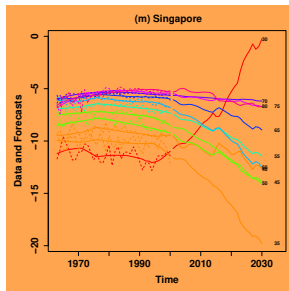
Lung cancer in Males, Singapore

Least Squares

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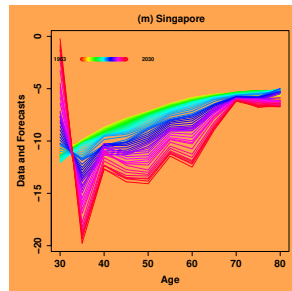
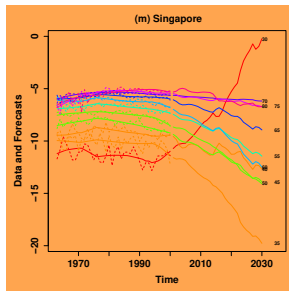
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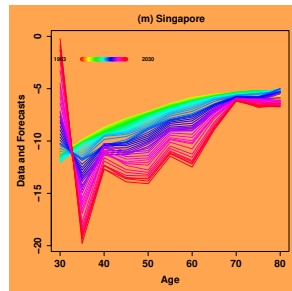
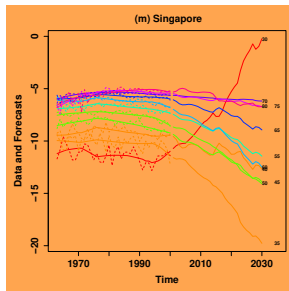
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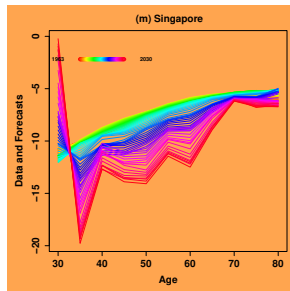
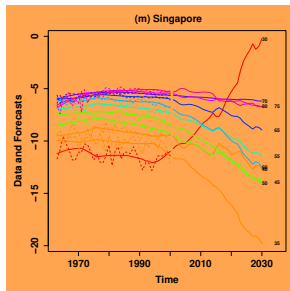


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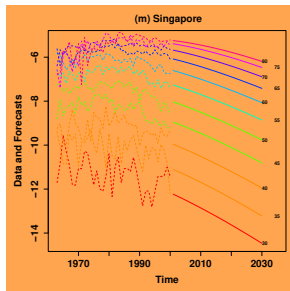
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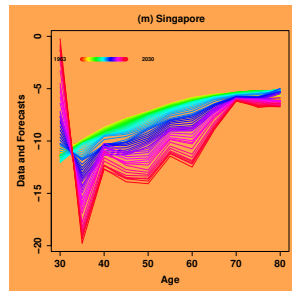
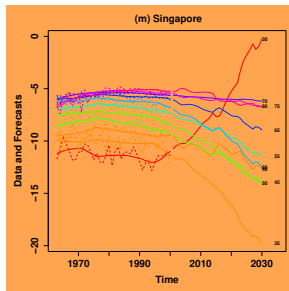
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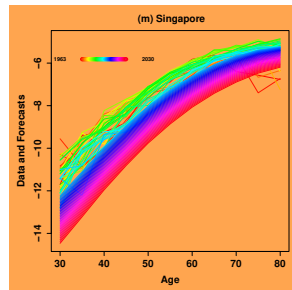
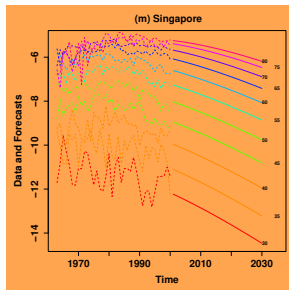
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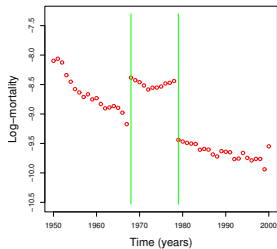


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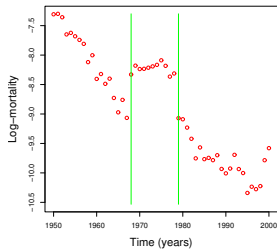


What about ICD Changes?

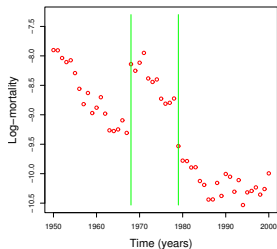
Other Infectious Diseases : USA , age 0 (m)



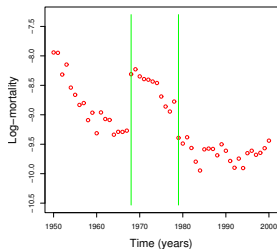
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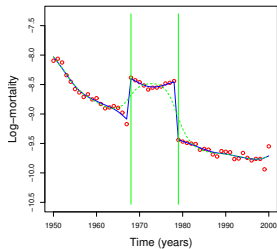


Other Infectious Diseases : United Kingdom , age 0 (m)

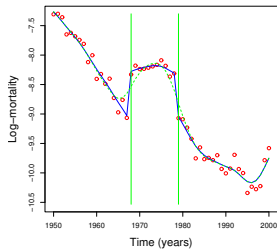


Fixing ICD Changes

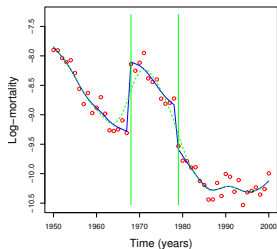
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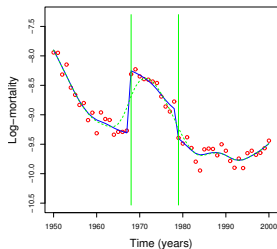
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A book manuscript, YourCast software, etc.

<http://GKing.Harvard.edu>

Preview of Results: Out-of-Sample Evaluation

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Mean Absolute Error in Males (over age and country)

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	<u>% Improvement</u>	
	Over Best Previous	to Best Conceivable
Cardiovascular	22	49
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where we have defined:

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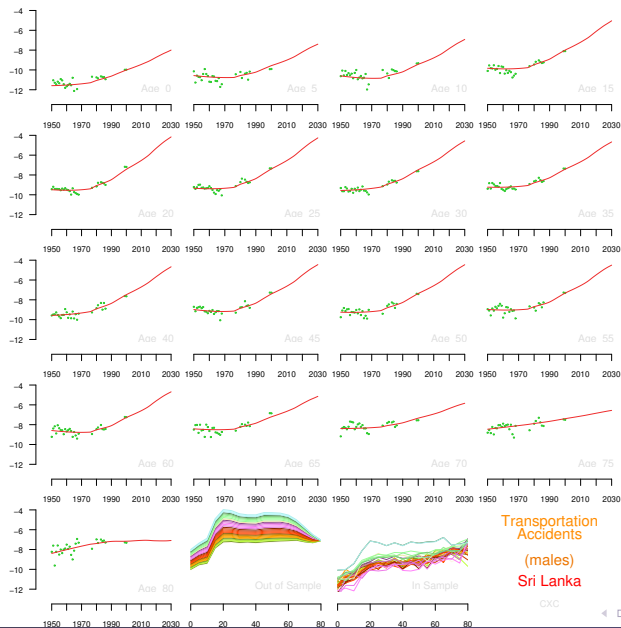
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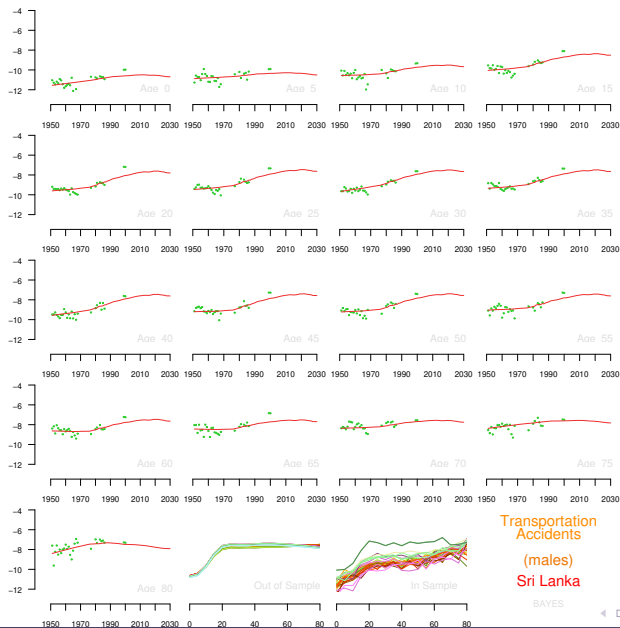
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Without Country Smoothing



With Country Smoothing



()

Demographic Forecasting

95 / 100

Formalizing Similarity

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Standard Bayesian Approach

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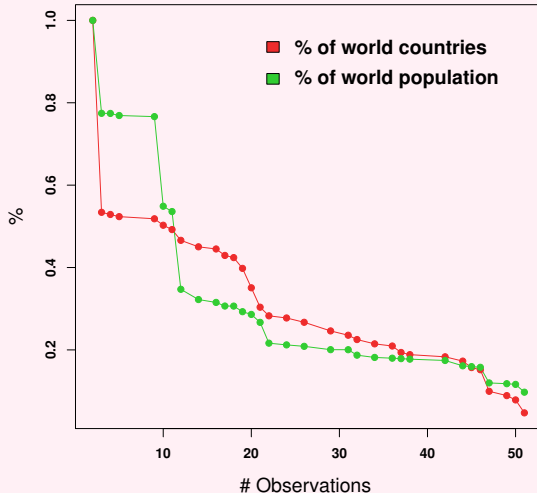
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Many Short Time Series

Coverage of WHO data base (age specific, all causes)



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Cardiovascular	0.34	0.27	0.19	22	49
Lung Cancer	0.36	0.27	0.17	24	47
Transportation	0.37	0.31	0.18	16	31
Respiratory Chronic	0.45	0.39	0.26	13	30
Other Infectious	0.55	0.48	0.32	12	30
Stomach Cancer	0.30	0.27	0.20	8	24
All-Cause	0.17	0.15	0.08	12	22
Suicide	0.31	0.29	0.18	7	17
Respiratory Infectious	0.49	0.47	0.28	3	7

Preview of Results: Out-of-Sample Evaluation

Mean Absolute Error in Males (over age and country)

	Mean Absolute Error			% Improvement	
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- Each row averages 6,800 forecast errors (17 age groups, 40 countries, and 10 out-of-sample years).

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- % to best conceivable = % of the way our method takes us from the best existing to the best conceivable forecast.
- The new method out-performs with the same covariates, for most countries, causes, sexes, and age groups.
- Does much better with better covariates